

Dense Irrational Periodicity with a Single Rational Exception

There is an analytic Hamiltonian brake system in which a dense but measure-zero set of irrational parameters is periodic, exactly one rational parameter is periodic, and every other rational parameter is nonperiodic.

For $0 < u < 1$, consider

$$H_u = \frac{1}{2}(p_x^2 + x^2) + \frac{1}{2}(p_y^2 + \Omega(u)^2 y^2),$$

where

$$\Omega(u) = 2 + u^2 + \sqrt{2}u \left(u - \frac{1}{2}\right).$$

Start every orbit from the same brake state,

$$x(0) = y(0) = 1, \quad p_x(0) = p_y(0) = 0.$$

The resulting motion is

$$x(t) = \cos t, \quad y(t) = \cos(\Omega(u)t).$$

The full orbit is periodic—equivalently, it reaches a second brake—exactly when

$$\Omega(u) \in \mathbb{Q}.$$

Almost no rational parameters work

Suppose $u \in \mathbb{Q}$. Then $2 + u^2$ is rational, whereas

$$\sqrt{2}u \left(u - \frac{1}{2}\right)$$

is irrational unless its rational coefficient vanishes. Therefore

$$\Omega(u) \in \mathbb{Q} \iff u \left(u - \frac{1}{2}\right) = 0.$$

Since $0 < u < 1$, the only rational periodic parameter is

$$\boxed{u = \frac{1}{2}}.$$

At that parameter,

$$\Omega\left(\frac{1}{2}\right) = \frac{9}{4},$$

so the orbit is genuinely periodic. Every other rational parameter in the interval produces quasiperiodic motion.

Nevertheless, irrational periodic parameters are dense

Set

$$u_* = 1 - \frac{1}{\sqrt{2}} \approx 0.292893218.$$

This parameter is irrational, but direct substitution gives

$$\Omega(u_*) = 2.$$

It therefore generates the elementary 1:2 periodic orbit

$$x(t) = \cos t, \quad y(t) = \cos(2t).$$

Notably, this explicit irrational periodic parameter lies close to the $u \approx 0.29$ region considered in the Pythagorean–Burrau investigation.

The complete periodic set is

$$\mathcal{P} = \Omega^{-1}(\mathbb{Q}).$$

Because Ω is nonconstant and continuous, the image of every open parameter interval contains an interval and hence a rational number. Thus $\mathcal{P}$ is dense in $(0, 1)$. On the other hand, every rational frequency has at most two preimages under the quadratic function Ω . Consequently, $\mathcal{P}$ is countable and has Lebesgue measure zero.

Removing its sole rational member leaves

$$\mathcal{P} \setminus \left\{ \frac{1}{2} \right\},$$

a dense, countable, measure-zero set consisting entirely of irrational periodic parameters.

Periodic irrational parameters are dense but measure zero; exactly one rational parameter is periodic; every other rational parameter is nonperiodic.

The example is deliberately engineered, but it shows exactly why measure, density, and a generic dimension count cannot determine which rational parameters lie on a periodic locus. An exact structural or arithmetic bridge is indispensable in the Pythagorean–Burrau problem.